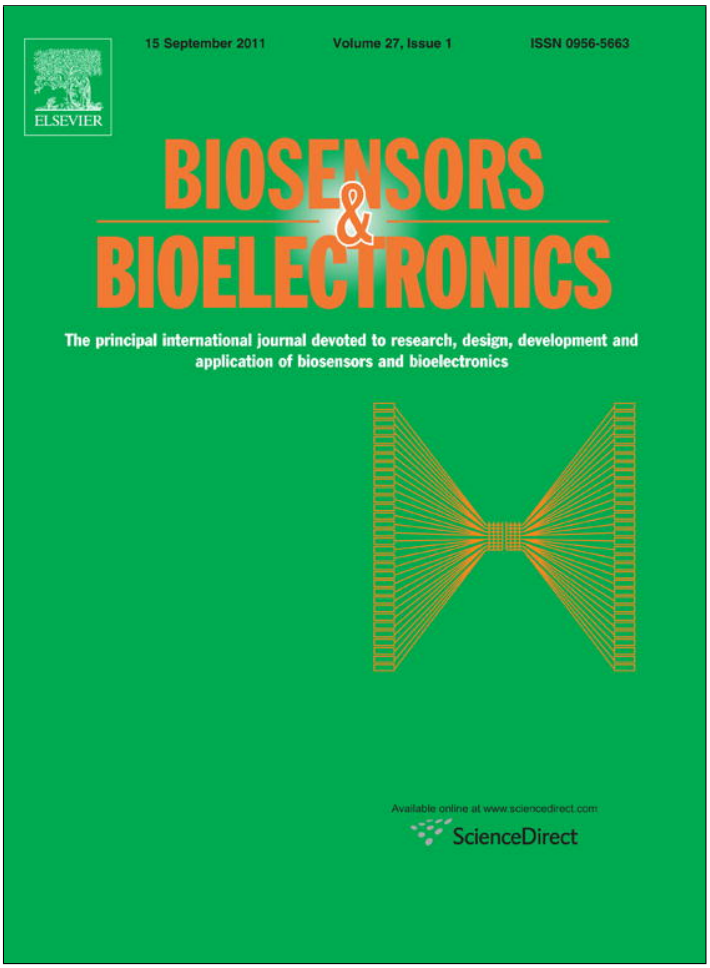
Provided for non-commercial research and education use.
Not for reproduction, distribution or commercial use.



This article appeared in a journal published by Elsevier. The attached copy is furnished to the author for internal non-commercial research and education use, including for instruction at the authors institution and sharing with colleagues.

Other uses, including reproduction and distribution, or selling or licensing copies, or posting to personal, institutional or third party websites are prohibited.

In most cases authors are permitted to post their version of the article (e.g. in Word or Tex form) to their personal website or institutional repository. Authors requiring further information regarding Elsevier's archiving and manuscript policies are encouraged to visit:

http://www.elsevier.com/copyright

Author's personal copy

Biosensors and Bioelectronics 27 (2011) 183–186

Contents lists available at ScienceDirect

Biosensors and Bioelectronics

journal homepage: www.elsevier.com/locate/bios

Short communication

A three-dimensional gold nanodendrite network porous structure and its application for an electrochemical sensing

Tran Ngoc Huan [a], Thothadri Ganesh [a], Kwang Soo Kim [b], Saetbyeol Kim [a], Sung-Hwan Han [a], Hoeil Chung [a,*]

[a] Department of Chemistry, Research Institute for Convergence of Basic sciences, Hanyang University, Seongdong-Gu, Haengdang-Dong, Seoul 133-791, Republic of Korea
[b] Samsung Electro-Mechanics, Suwon, Republic of Korea

ARTICLE INFO

Article history:
Received 20 April 2011
Received in revised form 6 June 2011
Accepted 10 June 2011
Available online 17 June 2011

Keywords:
Gold nanodendrite
Nanodendrite network porous structure
Nucleation
Arsenic detection

ABSTRACT

A three dimensional (3D) gold (Au) nanodendrite network porous structure constructed by a simple electrochemical synthetic method has been presented, and its utility for sensitive electrochemical measurement was demonstrated in this study. The 3D nanodendrite network porous structure was constructed on a platinum surface through electrodeposition of Au under the presence of hydrogen bubbles generated from the same surface. Iodide, used as a co-reagent, played an important role in the construction of the nanodendrite network by preventing continual growth of Au into larger agglomerates as well as inhibiting coalescence of neighboring nanodendrites. An electrochemical sensor incorporating the structure was built and used to detect As(III) in ultra low concentration range. For the purpose of comparison, bare gold and gold nanoparticle-incorporated electrodes were also prepared. With the use of 3D nanodendrite network porous structure, a much more sensitive detection of As(III) was possible due to its large surface area.

© 2011 Elsevier B.V. All rights reserved.

1. Introduction

The design and synthesis of nano-structures on electrode surface have been recently studied to improve sensitivity of electrochemical measurement by increasing surface area for analysis (Juste et al., 2005; Guo and Wang, 2007). The incorporation of nano-structures, such as nanoparticles, nanorods, nanotubes and nanodendrites, onto electrode surface has been largely considered for diverse analytical applications. Among them, gold (Au) nanodendrites have drawn a great attention due to availability of large surface area within a structure. Until now, to build nanodendrite structure, either chemical synthesis or electrochemical synthesis has been demonstrated. Chemical synthesis of Au nanodendrite mostly involves the reduction of Au salt or chloroauric acid in aqueous solutions in the presence of diverse reducing agents such as 3,4-ethylenedioxythiophene (EDOT) (Lu et al., 2007), hydroxylamine (NH_2OH) (Jasuja and Berry, 2009), a zinc plate and 1-butyl-3 methylimidazolium hexafluorophosphate [BMIM][PF_6] (Qin et al., 2008), and a mixture of dodecyltrimethylammonium bromide (DTAB) and β-cyclodextrin (β-CD) (Huang et al., 2010). In these reactions, Au nanodendrite structure is formed through the continual nucleation of Au. However, the construction of nanodendrite structures through chemical reactions requires long period of time and extensive use of chemicals. In comparison with chemical synthesis, electrochemical synthesis of Au dendrite is much faster and simpler without extensive use of reagents. Very recently, 2 different electrochemical strategies based on the electrodeposition of Au directly to the surface of electrode have been reported. First, the structure was built by applying solely negative potential, and applied for a DNA biosensor (Xu et al., 2010; Li et al., 2011). Second, the reduction of Au was performed under presence of cysteine (Lin et al., 2011). Adsorption of cysteine on Au helped to construct a nanodendrite structure. So far, the demonstrated nanodendrites constructed by either chemical or electrochemical synthetic ways are mostly two dimensional (2D) structures, formed in several localized areas or spread on a surface. To substantially increase surface area available for electrochemical measurements, rather than 2D structures, a three dimensional (3D) structure in which each nanodendrite is networked together will be highly beneficial. This useful nanodendrite-networked 3D structure has not been demonstrated so far.

In this article, we present a novel 3D Au nanodendrite network structure prepared by a simple and fast electrochemical synthesis. The basic strategy for synthesis of the structure was the combination of 2 simultaneously occurring electrochemical reactions: electrodeposition of Au onto a platinum surface in presence of iodide as a co-reagent to drive continual generation of nanodendrites and the simultaneous release of a hydrogen bubble at the

* Corresponding author. Tel.: +82 2 2220 0937; fax: +82 2 2299 0762.
E-mail address: hoeil@hanyang.ac.kr (H. Chung).

0956-5663/$ – see front matter © 2011 Elsevier B.V. All rights reserved.
doi:10.1016/j.bios.2011.06.011

Author's personal copy

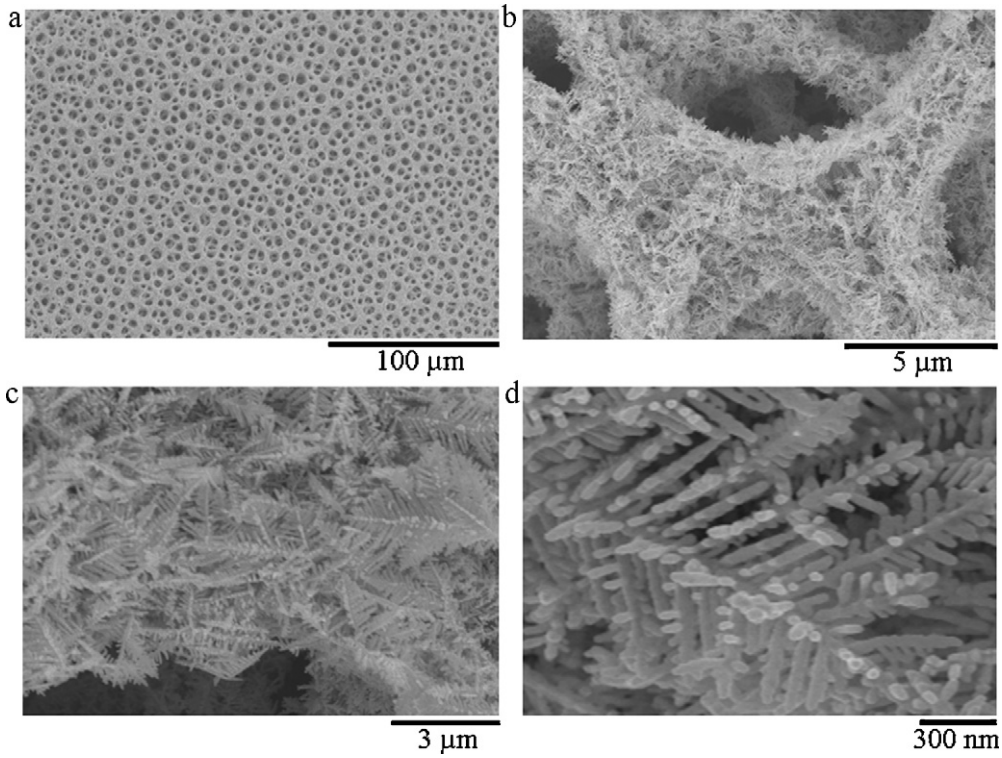


Fig. 1. SEM images of 3D Au nanodendrite network porous structure with 4 different fields of view.

surface to create pores (~5–10 μm) within the network structure. The purpose of pore generation inside the 3D structure was for facile access of analytes for electrochemical measurement. Using the 3D nanodendrite network porous structure, electrochemical detection of As(III) was performed. To compare the sensitivity of measurement, bare Au and Au nanoparticle incorporated electrodes were also prepared and used for the same measurement.

2. Experimental

2.1. Chemicals and instrumentations

Gold (III) chloride hydrate (99.999%), potassium iodide, sulfuric acid, hydrochloric acid and arsenic (III) oxide were purchased from Sigma–Aldrich. The electrochemical synthesis of 3D Au nanodendrite network porous structure was performed on a galvanostat (Epsilon, Bioanalytical Chemistry Systems Inc., USA).

All electrochemical measurements were performed at room temperature using a standard cell with a 3 electrode arrangement in which Ag/AgCl and Pt wire were used as reference and counter electrodes, respectively. The potential measurement was performed relative to the Ag/AgCl reference electrode, and all data were recorded using a potentiostat (eDAQ, Model PowerLab/4SP, Denistone East, NSW 2112, Australia).

2.2. The preparation of 3D Au nanodendrite network porous structure

For the construction of 3D Au nanodendrite network porous structure, a solution of 20 mM $HAuCl_4$, 0.5 M H_2SO_4, 1.0 mM KI and 2.5 M NH_4Cl was initially prepared. Then, a platinum electrode was immersed into the mixture solution and a current of 50 mA was applied over 140 s using a galvanostat (Epsilon, Bioanalytical Chemistry Systems Inc., USA). Finally, SEM images were taken on a Hitachi S-4800 SEM.

3. Results and discussion

3.1. Construction of 3D Au nanodendrite network porous structure

Fig. 1a shows the SEM images of the obtained 3D Au nanodendrite network porous structure with the largest field of view (240 μm × 160 μm). Pores at the surface as well as within the structure are clearly observable, and their size ranges from 5 μm to 10 μm. Overall the porous structure was uniform. Fig. 1b and c shows the magnified SEM images of 2 different locations on the structure with the field of view of 15 μm × 10 μm and 12 μm × 8 μm, respectively. The three-dimensional porous structure and the presence of many pores within the structure were clearly confirmed again, and the frame of the nanodendrite network is observable in these images. Fig. 1d shows a further magnified image of the nanodendrite network frame (field of view: 1.8 μm × 1.2 μm), in which a well ordered nanodendrite structure is obvious. Within the nanodendrite network frame, considerable void space is available and provides ultra high surface area for analytes to interact with the Au surface for further analytical applications.

Iodide plays an important role in the prevention of continual growth of Au particles as well as inhibiting particle aggregation (El-Deab et al., 2005a,b). At early stage of electrodeposition, iodide ions were instantaneously adsorbed onto the surfaces of newly generated Au crystals, producing a negative charge on their surface (El-Deab et al., 2005a,b; Chen et al., 1999; Shi and Paul, 2008). The repulsion among the negatively charged particles increased the rate of nucleation greater than that of particle growth, so formation of the nanodendrite structure was preferable to larger particle growth

Author's personal copy

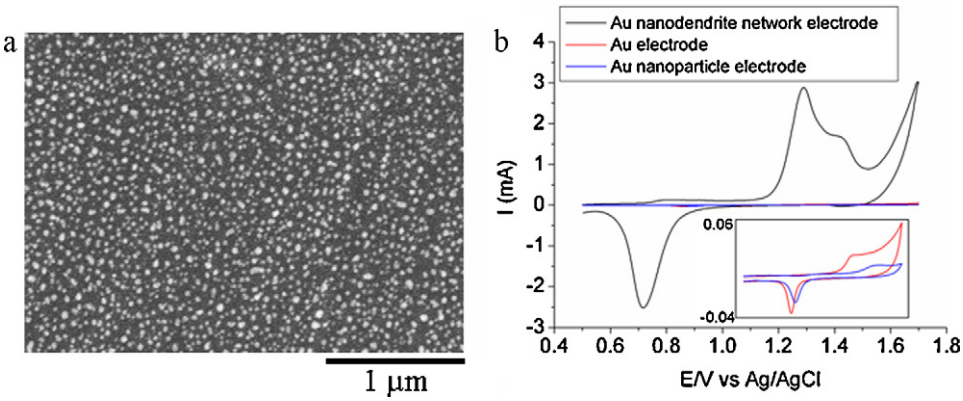


Fig. 2. SEM image of Au nanoparticle electrode (a) and cyclic voltammograms acquired from 3 electrodes in 0.5 M H_2SO_4 (b).

or aggregation. Further, in the presence of iodide, the negatively charged nanodendrites continued to network themselves in both lateral and axial directions, without coalescence of neighboring nanodendrites. Also, the use of NH_4Cl as a co-reagent increased the rate of hydrogen bubble generation (Cherevko et al., 2010), leading to a uniform porous structure and fostering the continual growth of the network structure by hindering of dendrite aggregation. To verify the role of iodide during electrodeposition, the structure was synthesized without iodide. The porous structure was partially constructed without uniformity and nanodendrite network was not clearly observed (refer to supplementary data).

3.2. Characterization of 3D Au nanodendrite network porous structure

To characterize the 3D Au nanodendrite network structure for an actual electrochemical measurement, diffusional electrode area based on Randles–Sevcik equation (Ressine et al., 2010) as below was calculated by acquiring cyclic voltammogram of a compound with known diffusion coefficient, $K_3[Fe(CN)_6]$ in this study. In reality, the estimation of effective surface area for analysis by considering diffusion of an analyte could be necessary since it has 3D structure.

$$i_p = 2.69 \times 10^5 n^{3/2} D^{1/2} A_{\mathrm{diff}} C \nu^{1/2} \quad (1)$$

Here, i_p is the peak current corresponding to reduction of redox species (Fe^{3+}/Fe^{2+}), n is the number of electron transfer, D is the diffusion coefficient of an analyte, A_{diff} is the diffusional electrode area, C is the molar concentration of an analyte and ν is the scan rate ($\mathrm{V\,s^{-1}}$). The C and D of $K_3[Fe(CN)_6]$ are 5×10^{-6} M and $7.5 \times 10^{-6}\,\mathrm{cm^2\,s^{-1}}$, respectively. In addition, a bare gold electrode was prepared and the corresponding diffusional electrode area was also calculated for the purpose of comparison. The CVs acquired from both electrodes are shown in supplementary data. The calculated diffusional electrode areas for the bare Au and 3D Au nanodendrite network electrodes were 0.008 and 0.261 $\mathrm{cm^2}$, respectively. The 3D Au nanodendrite network structure provides approximately 33 times larger diffusional area compared to that of bare Au surface, when $K_3[Fe(CN)_6]$ is measured.

To demonstrate analytical utility of the 3D Au nanodendrite network porous structure, the detection of As(III) was performed using the structure and corresponding analytical sensitivity was examined. For the comparative study, a bare Au electrode and Au nanoparticle-incorporated glassy carbon electrode (GCE) were also prepared for the same measurement. The Au nanoparticle-incorporated GCE (referred to as Au nanoparticle electrode), of the type which showed improved sensitivity for electrochemical measurement in several previous publications (Guo and Wang, 2007; Dai et al., 2004), was prepared using chronoamperometry from a solution containing 1 mM $NaAuCl_4$, 0.1 mM KI and 0.5 M H_2SO_4. All 3 electrodes used in the experiment had the same diameter of 1 mm.

Fig. 2a shows the surface SEM image of the Au nanoparticle electrode in which Au nanoparticles were well distributed on the GCE in sizes ranging from 30 nm to 60 nm. Fig. 2b shows cyclic voltammograms (CVs) obtained from the 3 electrodes in 0.5 M H_2SO_4 over a scan range from 0.5 V to 1.7 V. The CVs acquired from the bare Au and Au nanoparticle electrodes are magnified in the inset because the corresponding reduction peak intensities were much smaller than that from the Au nanodendrite network electrode. The y-span of the inset plot is only 0.1 mA. As observed, the reduction peak obtained from the Au nanodendrite network electrode was much higher than those from the other electrodes, indicative of the ultra high surface area within the structure.

3.3. Detection of As(III) using nanodendrite network structure

The sensitivities of 3 electrodes were examined by measuring a sample of 1 ppm As(III) in a 0.2 M HCl solution. For the measurement, differential stripping pulse voltammetry was used over a scan range from −400 mV to 500 mV. Stripping potential and scan rate were −400 mV and 100 $\mathrm{mV\,s^{-1}}$, respectively. During stripping, the sample solution was stirred at a constant rate. Fig. 3a shows the resulting voltammograms acquired from the 3 different electrodes. The peaks of As(III) were observed in all 3 cases; however, the peak intensity acquired from the Au nanodendrite network electrode was approximately 40 times higher than that from the Au nanoparticle electrode.

Measurement in a lower concentration range (0.1–70 ppb) was attempted using only the Au nanodendrite network electrode. To also evaluate sensor-to-sensor reproducibility, 4 Au nanodendrite network electrodes were separately prepared and individually measured at a given concentration. Fig. 3b shows the resulting voltammograms acquired from different concentrations of As(III) solutions. The voltammograms obtained from only 1 of the 4 replicate sensors are displayed. The reduction peak corresponding to 0.1 ppb is very small due to the large range of the y-scale; after magnification, the peak can be clearly observed with good signal-to-noise ratio.

The peak intensity clearly increased with the increase in As(III) concentration. The calibration curve generated using the corresponding peak heights from 4 individual measurements is also shown in the figure, and the response was linear over the tested concentration range. In addition, the peak heights obtained from

Author's personal copy

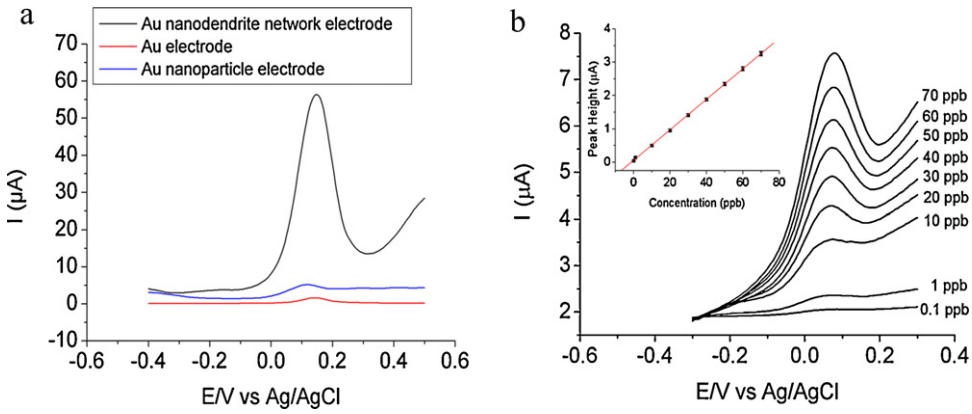


Fig. 3. Voltammograms acquired in a solution of 1 ppm As(III) using 3 different electrodes (a) and voltammograms acquired from different concentration of As(III) solution using the Au nanodendrite network electrode. The calibration curve constructed using the corresponding peak heights is also shown (b).

4 separate sensors at each concentration were similar, indicating superior sensor-to-sensor reproducibility. This is further evidence for the reproducible construction of the 3D Au nanodendrite network porous structure. The response was analyzed using regression analysis, and the resulting slope (sensitivity) and correlation coefficient (R^2) were $0.046 \pm 4.2 \times 10^{-5}$ μA ppb^{-1} and 0.998, respectively.

4. Conclusions

The Au nanodendrite network porous structure demonstrated in this study should be further valuable for sensitive and selective detection of diverse biomolecules when a target-specific sensing layer is formed on the surface of structure. In addition, the structure would be also applicable for surface enhanced Raman scattering (SERS)-based measurements since hot spots within the Au nanodendrites and their network would be largely available (Qiu et al., 2009; He et al., 2010). Future research will include the evaluation of the merits of the structure for SERS based nano-bio sensing.

Acknowledgements

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology (200900000000896).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.bios.2011.06.011.

References

Chen, A., Shi, Z., Bizzotto, D., Lipkowski, J., Pettinger, B., Bilger, C., 1999. J. Electroanal. Chem. 467, 342–353.
Cherevko, S., Xing, X., Chung, C.H., 2010. Electrochem. Commun. 12, 467–470.
Dai, X., Nekrassova, O., Hyde, M.E., Compton, R.G., 2004. Anal. Chem. 76, 5924–5929.
El-Deab, M.S., Sotomura, T., Ohsaka, T., 2005a. J. Electrochem. Soc. 152, C730–C737.
El-Deab, M.S., Sotomura, T., Ohsaka, T., 2005b. J. Electrochem. Soc. 152, C1–C6.
Guo, S., Wang, E., 2007. Anal. Chim. Acta 598, 181–192.
Huang, T., Meng, F., Qi, L., 2010. Langmuir 26, 7582–7589.
He, L., Lin, M., Li, H., Kim, N.J., 2010. J. Raman Spectrosc. 41, 739–744.
Jasuja, K., Berry, V., 2009. ACS Nano 3, 2358–2366.
Juste, J.P., Santos, I.P., Marzan, L.M.L., Mulvaney, P., 2005. Coord. Chem. Rev. 249, 1870–1901.
Li, F., Han, X., Liu, S., 2011. Biosens. Bioelectron. 5, 2619–2625.
Lin, T.H., Lin, C.W., Liu, H.H., Sheu, J.T., Hung, W.H., 2011. Chem. Commun. 7, 2044–2046.
Lu, G., Li, C., Shi, G., 2007. Chem. Mater. 19, 3433–3440.
Qin, Y., Sun, N., Zhao, N., Li, M., Qi, L., 2008. Chem. Mater. 20, 3965–3972.
Qiu, T., Zhou, Y., Li, J., Zhang, W., Lang, X., Cui, T., Chu, P.K., 2009. J. Phys. D: Appl. Phys. 42, 175403.
Ressine, A., Dominguez, C.V., Fernandez, V.M., Lacey, A.L.D., Laurell, T., Ruzgas, T., Shleev, S., 2010. Biosens. Bioelectron. 5, 1001–1007.
Shi, P., Paul, W.B., 2008. ACS Nano 2, 1581–1588.
Xu, X., Jia, J., Yang, X., Dong, S., 2010. Langmuir 26, 7627–7631.